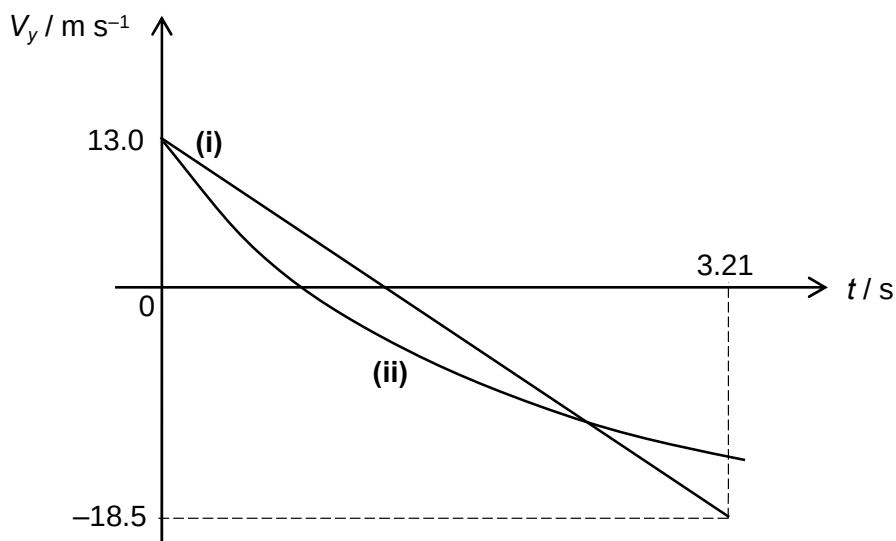


- 1 (a) (i) $-8.75 = (15.0 \sin 60^\circ) t + \frac{1}{2} (-9.81) t^2$ M1
 $8.75 = (-15.0 \sin 60^\circ) t + \frac{1}{2} (9.81) t^2$ (M1)
- solving $t = 3.21$ s A0
- (ii) range of projectile = $(15.0 \cos 60^\circ) (3.21) = 24.075$ m M1
- distance travelled by ship = $0.450 \times 3.21 = 1.4445$ m M1
- $D = 24.075 + 1.4445 = 25.5$ m A1
- (b) horizontal component of air resistance causes horizontal deceleration M1
- so range of projectile shorter and D will be shorter A1

(c)



- (i) straight line with negative gradient with final speed $>$ original speed M1
- label initial speed on y -axis and time of flight on x -axis A1
- (ii) correct trend for graph (ii) M1
- velocity = 0 at time earlier than (i) A1

